PROBLEM FS5

Gate between Two Liquids

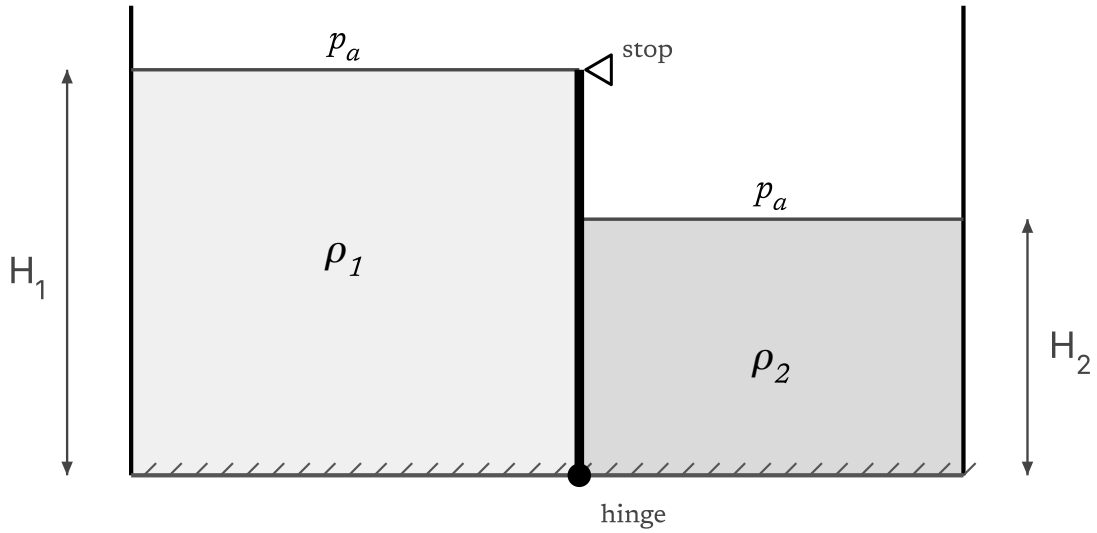
Original statement · fluid_statics_5.pdf

Plane surface | Two liquids | Moment about a hinge | Critical level

OFFICIAL PROBLEM STATEMENT

The gate of the attached figure separates two liquids of densities ρ_1 and $\rho_2 > \rho_1$. The height of the liquid on the two sides of the gate is H_1 and $H_2 < H_1$. The gate is hinged at its lower end and rests on a stop at its upper end, as shown in the attached figure, so that the configuration is in equilibrium as long as H_2 stays below a certain critical value, H_c, to be determined as a function of H_1, ρ_1 and ρ_2.

Translated from the Spanish original. The sheet gives no solution for this problem.



Statement figure (redrawn from fluid_statics_5.pdf, p. 1): gate of height H_1 between the liquids ρ_1 (left, height H_1) and $\rho_2 > \rho_1$ (right, height $H_2 < H_1$), hinged at the bottom and resting on a stop at the top

Source: fluid_statics_5.pdf (page 1, sheet of 11-2-00). Notes.pdf, Chapter 5: Eqs. 5.10 and 5.11.

Worked solution

Hypotheses, Data and Unknowns

- Both liquids at rest, incompressible, immiscible and separated by the gate; air at p_a above both, so only gauge pressures count.
- The gate is a vertical rigid plate of height H_1 and width b (normal to the paper), hinged at the bottom (point O) without friction; its weight passes through the hinge when it is vertical and has no moment about O.
- The stop at the top is on the side of liquid 2: it can only push the gate towards liquid 1, never pull.
- Data: ρ_1, ρ_2, H_1, H_2, g, b. Unknowns: the critical level H_c (and, as a by-product, the force on the stop).

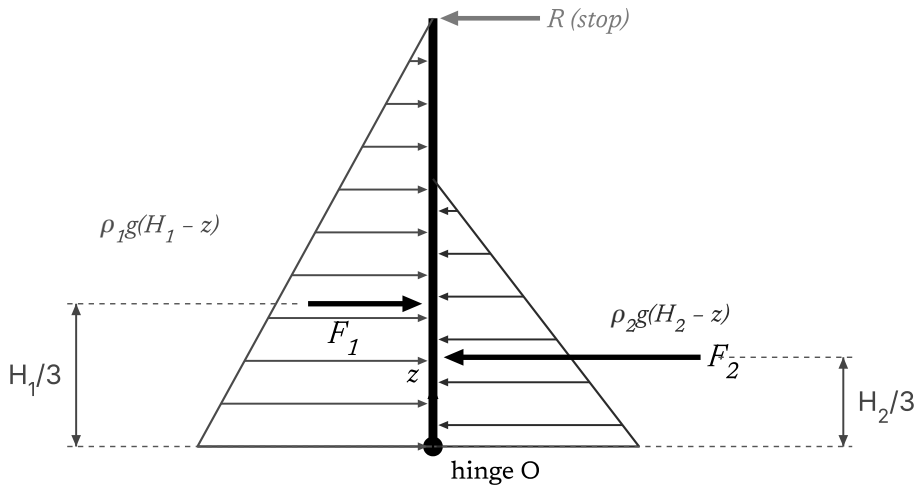
Symbol	Meaning	SI unit
ρ_1, ρ_2	densities, $\rho_2 > \rho_1$	kg/m^3
H_1, H_2	liquid heights	m
b	width of the gate	m

Coordinates and the Pressure on Each Face

Take z vertical, pointing up from the hinge, so the gate covers $0 \le z \le H_1$. By (5.10), measured from each free surface:

$$p_1(z) - p_a = \rho_1 g (H_1 - z), \quad 0 \le z \le H_1; \qquad p_2(z) - p_a = \rho_2 g (H_2 - z), \quad 0 \le z \le H_2$$

The strip between z and $z + dz$ has the area $b\,dz$, its force is horizontal, and its arm about the hinge is z. Liquid 1 pushes the gate towards liquid 2 (normal towards liquid 1 is $-\vec{e}_x$), liquid 2 pushes it back.



Free-body diagram of the gate (unit width): the triangular gauge pressures $\rho_1 g(H_1 - z)$ and $\rho_2 g(H_2 - z)$, their resultants F_1 and F_2 at $H_1/3$ and $H_2/3$ above the hinge O, and the stop reaction R at the top

Moments about the Hinge

Moment of liquid 1 (turning the gate towards liquid 2):

$$M_1 = \int_0^{H_1} \rho_1 g(H_1 - z)\, z\, b\, dz = \rho_1 g b \left[\frac{H_1 z^2}{2} - \frac{z^3}{3}\right]_0^{H_1} = \rho_1 g b \left(\frac{H_1^3}{2} - \frac{H_1^3}{3}\right) = \frac{\rho_1 g b H_1^3}{6}$$

Moment of liquid 2 (turning it back), with the same integral and H_2 in place of H_1:

$$M_2 = \int_0^{H_2} \rho_2 g(H_2 - z)\, z\, b\, dz = \frac{\rho_2 g b H_2^3}{6}$$

Check with the resultants: each triangular distribution has the resultant $\rho_i g b H_i^2/2$ applied at its centroid, $H_i/3$ above the bottom, and $(\rho_i g b H_i^2/2)(H_i/3) = \rho_i g b H_i^3/6$.

Equilibrium and the Critical Level

Moment balance about the hinge, with the stop reaction R (pushing towards liquid 1) acting at the top, arm H_1:

$$M_1 - M_2 - R\,H_1 = 0 \quad \Longrightarrow \quad R = \frac{gb\,(\rho_1 H_1^3 - \rho_2 H_2^3)}{6H_1}$$

The stop can only push, $R \geq 0$, so the configuration is in equilibrium while $\rho_1 H_1^3 \geq \rho_2 H_2^3$. The critical level is the one with $R = 0$:

$$\rho_1 H_1^3 = \rho_2 H_c^3 \quad \Longrightarrow \quad H_c = H_1 \left(\frac{\rho_1}{\rho_2}\right)^{1/3}$$

For $H_2 > H_c$ the moment of liquid 2 wins, the gate leaves the stop and turns towards liquid 1.

Answers and Checks

- $H_c = H_1(\rho_1/\rho_2)^{1/3}$; equilibrium for $H_2 < H_c$; stop force $R = gb(\rho_1 H_1^3 - \rho_2 H_2^3)/(6H_1)$.
- **Consistency with the statement**: $\rho_2 > \rho_1$ gives $H_c < H_1$, so the critical level lies within the gate, as it must for $H_2 < H_1$.
- **Limiting case** $\rho_2 = \rho_1$: $H_c = H_1$, two equal columns of the same liquid balance exactly.
- **Not the force balance**: the horizontal forces are equal when $\rho_1 H_1^2 = \rho_2 H_2^2$, which gives a different level, $H_1(\rho_1/\rho_2)^{1/2}$; the gate turns about the hinge, so the moments decide (the difference is taken by the hinge reaction).
- **Dimensions**: H_c is H_1 times a dimensionless ratio, m; R is (m s^{-2})(m)(kg m^{-3} m^3)/m = N.
- **Check against the official key**: the sheet has no handwritten solution for this problem; the result was checked with SymPy (`tools/fluids_fs/verify_fluid_statics.py`).